Seikh Souvagya Mustakim

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Education

Parala Maharaja Engineering College **2023 – 2027**
B.Tech, Electronics & Telecommunications; CGPA: 8.39/10 *Berhampur, Odisha*
Nalanda Public School, Cuttack — Class XII (Science), 91% *2020 – 2022*

Research & Internship Experience

IASc Summer Research Fellow **Jun 2026 – Aug 2026**
Dept. of Electronic Systems Engineering, IISc Bengaluru *Bengaluru, India*
- Developed an automated PCB reverse-engineering pipeline using YOLO-based oriented object detection to localize surface-mount devices and physical I/O ports.
- Integrated a multimodal Vision LLM for robust optical character recognition, successfully extracting component part numbers from low-contrast integrated circuits.
- Built an end-to-end Flask web application to process hardware images and automatically generate structured Hardware Bill of Materials (HBoM).

Generative AI Intern **Jun 2025 – Aug 2025**
RMSoEE, IIT Kharagpur
- Completed a hands-on GenAI program covering Deep Learning fundamentals, Retrieval-Augmented Generation (RAG), and Model Context Protocol (MCP).
- Designed and deployed end-to-end generative AI workflows utilizing vector databases and prompt orchestration.

Technical Skills

- **Robotics:** ROS2 Humble, Gazebo, RViz2, RPLidar, SLAM Toolbox, Nav2, AMCL, A*, PID
- **Embedded:** RPi, Arduino, ESP32, MPU6050, Jetson Nano, Tiva C, Embedded Linux
- **CV & ML:** OpenCV, YOLOv8, MediaPipe, Lucas-Kanade, TensorRT, CNNs, OCR, Scikit-learn
- **GenAI:** RAG Pipelines, MCP, Prompt Engineering, Vector DBs
- **Languages:** Python, C, C++
- **Tools:** Git, Linux, VS Code, Jupyter, CCS, PlatformIO

Projects

- **Destination_bot — Autonomous Navigation on ROS2** GitHub
  ROS2 · Gazebo · SLAM Toolbox · Nav2 · Python
  - Designed a custom URDF differential-drive robot; integrated SLAM for mapping and a hand-rolled A* planner for collision-free autonomous navigation in simulation.
- **FemeVision — Multi-Modal Safety Perception** GitHub
  YOLOv8 · Lucas-Kanade · MediaPipe · OpenCV · Streamlit
  - Fused object detection, optical flow, and pose tracking into a real-time violence-detection pipeline — **Runner-up**, Hack for Tomorrow, VSSUT Burla 2025.
- **Face Recognition Attendance System** GitHub
  PyTorch · MTCNN · FaceNet · SVM · Flask
  - Built an end-to-end pipeline (MTCNN → FaceNet embeddings → SVM) served through a Flask web app for real-time attendance logging.
- **IMU-Coordinated Bot with Heading Stabilisation** GitHub
  MPU6050 · Madgwick Filter · PID Control · Arduino
  - Fused 6-DOF IMU data via Madgwick filter; implemented closed-loop PID heading stabilisation with auto-calibration and deadband compensation.

Leadership & Achievements

- **Vice Chair — IEEE EDS SBC, PMEC** *2026–27*
  Elected to lead technical events, industry talks, and student outreach for the Electron Devices Society chapter.
- **Coordinator, R&I Wing — Robotics Club, PMEC** *2025–26*
  Spearheaded research initiatives; organised RoboExpo-2k25, Robomania 2k26, and the Innovation Challenge Hackathon.
- **Semifinalist — Robowar 8 kg, IIT KGP Kshitij** *Jan 2026*
  Designed and competed with an 8 kg combat robot nationally, handling full embedded integration under competition constraints.
- **Runner-up — Hack for Tomorrow, VSSUT Burla** *Jan 2025*
  Led a 4-member team to build a real-time violence detection system using optical flow and Harris corner detection in 24 hrs.

Certifications

Autonomous Vehicles Bootcamp (Robolearn LLP) | Generative AI: Learn by Doing (IIT Kharagpur) | Embedded Systems (NIELIT)